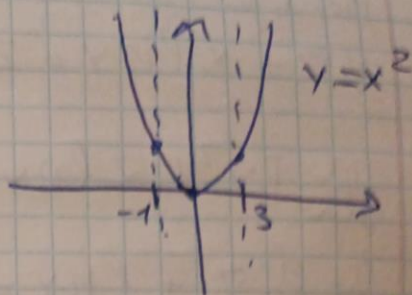


v 22 10 25 1

s 13 page ex 10

$$f(x, y) = \frac{\sqrt{y-x^2}}{1-x^2}$$



$$\sqrt{y-x^2} \Rightarrow y-x^2 \geq 0$$

defined  $y \geq x^2$

$$1-x^2 \Rightarrow 1-x^2 = 0$$

not defined  $x = \pm 1$

Domain of  $f$ :

$$(x, y) \mid y \geq x^2, x \neq \pm 1$$

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$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 y e^y}{x^4 + 4y^2} \Rightarrow f(x, y) = \frac{x^2 y e^y}{x^4 + 4y^2}$$

$$f(x, 0) = 0 \Rightarrow x \neq 0$$

$\Rightarrow x$ -axis is

$$f(x, y) \Rightarrow 0 \quad (x, y) \Rightarrow 0$$

$$(0, 0) \Rightarrow y\text{-axis or } y = x \text{ also } 0$$

$$f(x, x^2) = \frac{x^2 x^2 e^{x^2}}{x^4 + 4(x^2)^2} = \frac{x^4 e^{x^2}}{5x^4} = \frac{e^{x^2}}{5} \Rightarrow x \neq 0$$

$$f(x, y) \Rightarrow e^{y/5} = \frac{1}{5} \text{ as } (x, y) \Rightarrow (0, 0)$$

parabola  $y = x^2$

So limit doesn't exist

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$$yz + x \ln y = z^2$$

$$\frac{d}{dx} (yz + x \ln y) = \frac{d}{dx} (z^2)$$

$$y \frac{dz}{dx} + \ln y = 2z \frac{dz}{dx}$$

$$\ln y = 2z \frac{dz}{dx} - y \frac{dz}{dx}$$

$$\ln y = (2z - y) \frac{dz}{dx} \quad \frac{dz}{dx} = \frac{\ln y}{2z - y}$$

$$\frac{d^2}{y dx dy} + z \cdot 1 + x \cdot \frac{1}{y} = 2z \frac{dz}{dy}$$

$$z \neq \frac{x}{y} = 2z \frac{dz}{dy} - y \frac{dz}{dy}$$

$$\frac{dz}{dy} = \frac{z + \frac{x}{y}}{2z - y} = \frac{x + yz}{y(2z - y)}$$

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$$\sqrt{y + \cos^2 x} \approx 1 + \frac{1}{2} y$$

$$f(x, y) = \sqrt{y + \cos^2 x}$$

$$f(x, y) = \frac{1}{2} (y + \cos^2 x)^{-\frac{1}{2}} (\frac{1}{2} \cdot \cos x) (-\sin x) =$$

$$= -\frac{\cos x \sin x}{\sqrt{y + \cos^2 x}}$$



$$f_x(x, y) = \frac{1}{2}(y + \cos^2 x)^{-\frac{1}{2}}(1) = \frac{1}{2\sqrt{y + \cos^2 x}}$$

By Theorem 8:  $f_x$  &  $f_y$  are continuous functions for  $y > -\cos^2 x$ , so  $f$  is differentiable at  $(0, 0)$

$$f_x(0, 0) = 0 \quad f_y(0, 0) = \frac{1}{2}$$

$$f(x, y) \approx f(0, 0) + f_x(0, 0)(x - 0) + f_y(0, 0)(y - 0) = 1 + 0x + \frac{1}{2}y = 1 + \frac{1}{2}y$$

855 page ex 23

$$w = xy + yz + zx \quad x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = r\theta$$

$$\frac{dw}{dr}, \frac{dw}{d\theta} \quad \text{when } r = 2 / \theta = \frac{\pi}{2}$$

$$\frac{dw}{dr} = \frac{dw}{dx} \frac{dx}{dr} + \frac{dw}{dy} \frac{dy}{dr} + \frac{dw}{dz} \frac{dz}{dr}$$

$$= (y + 2)(\cos \theta) + (x + 2)(\sin \theta) + (y + x)\theta$$

$$\frac{dw}{d\theta} = \frac{dw}{dx} \frac{dx}{d\theta} + \frac{dw}{dy} \frac{dy}{d\theta} + \frac{dw}{dz} \frac{dz}{d\theta}$$

$$= (y+2)(-r \sin \theta) + (x+2)(2 \cos \theta) + (y+x)r$$

$$r=2 \quad \theta = \frac{\pi}{2} \Rightarrow x=0, y=2, z=\pi$$

$$\frac{dw}{dz} = (2+\pi) \cdot 0 + (0+\pi) \cdot 1 + (2+0) \frac{\pi}{2} + 2\pi$$

$$\frac{dw}{d\theta} = (2+\pi)(-2) + (0+\pi) \cdot 0 + (2+0) \cdot 2 = -2\pi$$